

William Stewart
163 Cuba Hill Rd.
Huntington, NY 11743
March 23, 2026

Attn: Marlene H. Dortch, Secretary
Federal Communications Commission
45 L Street NE
Washington, DC 20554

Re: Technical Rebuttal and Formal Opposition
File No. LOA-20260108-00016 (Space Exploration Holdings, LLC)

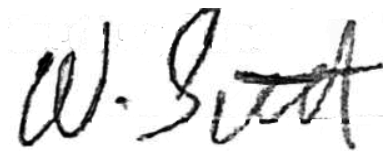
Dear Ms. Dortch,

Enclosed for filing in the above-referenced proceeding is a Technical Rebuttal to the Consolidated Opposition filed by Space Exploration Holdings, LLC ("SpaceX").

This filing specifically addresses the thermal impossibilities and thermodynamic contradictions inherent in the Applicant's proposed Gen3 orbital architecture. Furthermore, it highlights the lack of "Novelty" in the Applicant's claims due to significant Prior Art.

The Petitioner requests that the Commission take notice of these fundamental physical constraints, as the proposed system is incapable of safe operation as currently described.

Respectfully submitted,

A handwritten signature in black ink, appearing to read "W. Stewart", written over a light gray grid background.

/s/ William Stewart,
Petitioner

Technical Rebuttal: The Radiator Paradox and Thermal Impossibility

I. The "3-Kelvin" Deception

In its Consolidated Opposition [Section 2.6], SpaceX asserts that the high-density deployment of AI-processing satellites is sustainable due to the "natural 3-Kelvin cooling environment of Deep Space." This statement is a fundamental misrepresentation of thermodynamics.

While the cosmic microwave background temperature is approximately 2.7 K, space is a **vacuum**. In a vacuum, heat cannot be dissipated via conduction or convection. It can only be removed via **thermal radiation**.

II. The Stefan-Boltzmann Constraint

The rate of heat transfer from a satellite is governed by the Stefan-Boltzmann Law:

$$P = e * s * A * T^4$$

Where:

P = Power Radiated (Watts).

e = Emissivity; the efficiency of the radiator surface (0 to 1)

s = Stefan-Boltzmann Constant ($5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$)

A = Total Surface Area of the radiator.

T = Absolute temperature of the satellite (Kelvin).

To run the server-grade AI-processing clusters required for a '**Type II**' architecture as referenced in the Gen3 filing, each satellite must dissipate approximately **15–20 kW** of heat—matching the power density of terrestrial AI data centers. To maintain an operational temperature that doesn't melt the silicon (approx. **350 K**), a satellite would require a radiator surface area exceeding its own physical dimensions. It is important to note that this represents **the ideal case** (blackbody radiation), meaning the real-world performance will be even lower due to emissivity (e) being less than 1.

III. The "Baking" Effect of 1 Million Satellites

SpaceX's plan for 1 million satellites creates a "thermal shell" around the Earth. Because these satellites are in high-density clusters, they are not radiating into "empty 3K space"; they are radiating at **each other**.

When Satellite A radiates heat, Satellite B absorbs a portion of that radiation. At the proposed density, the effective "background temperature" is no longer 3 K—it rises significantly, reducing the efficiency of every radiator in the fleet. In coordination with independent scientists and astrophysicists, and upon rigorous review of these thermal phenomena, I have formalized this constraint which I have named the

"Radiator Paradox": the more satellites you add to increase "AI capacity," the less heat you can actually remove per unit, leading to a thermal runaway effect, inevitable hardware failure, and increased orbital debris from "cooked" electronics.

The physical reality of the Radiator Paradox is not merely a theoretical concern; it is a gap in the Applicant's own engineering baseline. In its Consolidated Opposition, the Applicant admits that 'no widely accepted standards currently exist' for infrared emissions, and that it is still 'developing predictive models' to evaluate these effects (See Opposition at 14).

This is a stunning admission of technical prematurity. The Applicant is requesting a license to populate a 2,000 km shell with one million high-heat processors while simultaneously admitting it lacks the standards or the finalized models to predict how that heat will propagate. In the vacuum of space, where the 3-Kelvin background acts as an insulator rather than a heatsink (The 3-Kelvin Deception), this lack of modeling is not a minor oversight; it is a guarantee of mission-wide thermal runaway, proven by SpaceX's own admission regarding the lack of infrared standards and predictive models. **SpaceX is essentially asking the FCC to be their "Alpha Tester" for a one million-satellite constellation.** In truth, by labeling this project an 'Alpha Test,' the Petitioner is being generous. Given the Applicant's admitted lack of predictive modeling, the Gen3 architecture remains in a 'Pre-Alpha' Engineering Validation Testing phase (EVT). It is unclear if the Commission's current technical staff has evaluated the system at this granular level of developmental maturity. Essentially, the Applicant is attempting to skip the foundational 'Proof of Concept' stage, which would determine if such heat dissipation is physically possible, and move directly to an irreversible global rollout (**except in SPACE**).

IV. The Valuation-Driven Narrative vs. Physical Reality

The Applicant's pursuit of a "Novelty" waiver under File No. LOA-20260108-00016 appears to be a prerequisite for their projected \$1.5 Trillion valuation. However, "Novelty" requires two things that are currently absent from the record: Originality & Feasibility. The Applicant's claims ignore existing Prior Art (Starcloud-1, 2025). The Applicant's rebuttal introduces a "3-Kelvin Cooling" model that is physically impossible at the proposed density. If the FCC grants a waiver based on a narrative that is both unoriginal and physically impossible, it effectively validates a \$750 billion valuation "pump" built on a foundation of engineering negligence.

V. Consolidated Opposition

The Applicant stated that it would respond to issues 'collectively', yet the Applicant's March 16th Consolidated Opposition attempts to dismiss the concerns of multiple petitioners with a single, scientifically flawed narrative, making their strategy appear to be one of collective dismissal.

VI. Unrebutted Technical & Physiological Concessions

The Applicant fails to provide a single word of rebuttal to critical technical, logistical, & health hazards raised in my Petition and Addendums. Therefore, under standard administrative procedure, the Applicant has effectively conceded these points as uncontested facts:

- A. The Twilight Extension Paradox:** The Applicant offers no rebuttal to the fact that at 2,000 km, the constellation will remain illuminated significantly longer into the night than existing systems, rendering traditional 'darkening' techniques ineffective against prolonged visibility.
- B. Cumulative Skyglow:** The Applicant addresses individual satellite brightness but remains silent on the aggregate diffuse luminosity of 1,000,000 objects, which creates a permanent 'Skyglow' that bypasses magnitude-based mitigations.
- C. Biological Impact:**
 - 1. Blue-Wavelength Suppression: As highlighted by recent warnings from Northwestern University sleep experts (March 2026), the specific spectrum of reflected sunlight from 1,000,000 metallic surfaces increases blue-rich light at night.
 - 2. Health Concession: The American Medical Association (AMA) and Northwestern physicians have documented that even low-level chronic exposure to artificial skyglow suppresses melatonin production, leading to sleep-wake disorders, metabolic irregularities, systemic circadian rhythm disruption, and weakened immune systems.
- D. The "Persistence Gap":** The Petitioner previously identified a critical Persistence Gap: The Applicant's thermal models rely on instantaneous cooling assumptions that ignore the cumulative heat-soak of 1,000,000 high-density processors operating in a vacuum, as corroborated by Gérard Scheitler in Section VII.
- E. Unchallenged Prior Art:** The Petitioner previously identified that the Applicant's system is not a novel architecture but is preceded by Starcloud-1 (2025). This Prior Art established the physical baseline for high-density, vacuum-integrated terrestrial hardware—the exact "Orbital Data Center" model the Applicant is now proposing. By failing to distinguish its architecture from Starcloud-1, the Applicant has failed to meet its burden of proof. The Commission cannot find that this system is "in the public interest" when the only existing technical baseline for such a system (Starcloud-1) resulted in the exact thermal and operational instabilities identified by the Petitioner in item D.
- F. The 0.8% Operational Tax:** The Applicant provides no modeling for the fuel sustainability of 1 million maneuver-capable units, essentially conceding that the 'Maneuver Paradox' remains an unsolved scalability failure. As constellation density increases, the fuel and "duty cycle" required for collision avoidance maneuvers grows. The probability of collisions grows at such a compounding, non-linear rate that it necessitates near-continuous maneuvering, eventually exceeding the propellant capacity and thermal recovery time of the hardware.
- G. The "Impossible Cadence" and Logistical Failure:** The Petitioner previously identified the Impossible Cadence required to maintain a 1,000,000-satellite constellation with a stated 5-

year operational lifespan. To achieve 50% within the timeframe (if no waiver is granted), the Applicant would be required to launch, deploy, and commission approximately **288 satellites every day, 365 days a year**. To maintain this population of 1M satellites thereafter, the Applicant would be required to launch, deploy, and commission approximately **548 satellites per day, 365 days a year, in perpetuity** merely to offset the natural orbital decay and operational expiration of the fleet. The Applicant's Opposition provides no logistical model, launch manifest, or historical precedent for such a cadence.

- H. **The Replacement Flux:** As noted by **Professor Hugh Lewis**, this necessitates an annual transit of **400,000 satellites** (200,000 launching and 200,000 de-orbiting) moving through LEO at any given time. The Applicant's failure to address this "Permanent Wall of Hardware" proves the application is based on a valuation-driven narrative rather than a physically achievable launch and disposal reality.
- I. **The 0.5% Failure Rate (The "Debris Factory"):** The Petitioner previously identified that even with the Applicant's best-case "success" metrics, a 0.5% hardware failure rate on a 1,000,000-unit constellation results in **5,000 "dead" satellites** orbiting as unguided kinetic projectiles at any given time. The Applicant's Opposition provides no mitigation strategy for the accumulation of 5,000 units of "dark" debris. As noted by **Professor Hugh Lewis**, this creates a "Debris Factory" in the most congested LEO shells. By failing to rebut the mathematical certainty of 5,000 annual failures, the Applicant concedes that its "Data Center" architecture carries an unmitigated risk of Kessler-Syndrome-level cascading collisions, rendering the "Public Interest" finding impossible.

VII. Expert Corroboration of Non-Linear Scalability Hazards:

- A. **Volumetric Fallacy:** As noted by Professor Hugh Lewis (University of Birmingham), the Applicant's "0.005%" occupancy claim is a mathematical deception that ignores the dynamic "safety tori" and the annual transit flux of 400,000 satellites (replacements/retirees), which effectively claims the entire LEO volume. 0.005% is still a massive 60 Million cubic kilometers, and Orbits aren't static volumes, they are paths.
- B. **Operational Resilience:** Gérard Scheitler (NATO/EU Space Ops veteran) has confirmed that the thermal constraints and "incident management" risks identified by the Petitioner are real operational considerations that require the very "verifiable data" the Applicant admits it lacks.
- C. **Fractured and Prejudged Record:** The Petitioner adopts by reference the findings of Nickolai Bakken (March 15/17) regarding the "internally irreconcilable" nature of the application (3 satellites described vs. 1 million requested) and the documented "Prejudgment" of the merits by the Commission's Chairman on social media platforms owned by the Applicant's principal.

VIII. Regulatory Integrity:

- A. Procedural and Ownership Defects:** The Applicant has failed to provide a consistent account of the system's ownership and control architecture. The record contains internally irreconcilable data—describing a system of 3 satellites in one section and 1,000,000 in another—rendering the application legally deficient and incapable of a "Public Interest" determination.
- B. Fractured Public Record:** As documented by Bakken, the public record in this proceeding is "fractured." Critical technical disclosures and "pocket filings" have been shielded from the public via the 7-day server blackout, depriving the Commission of the rigorous peer review required for a deployment of this scale.
- C. Use of "Pocket Filing" to Evade Technical Scrutiny:** The Petitioner previously identified the Applicant's use of a "Pocket Filing" tactic, filing at the extreme deadline of a government shutdown period, to minimize the window for technical peer review. The Applicant's Opposition fails to provide any rebuttal or justification for this filing timeline. This confirms a pattern of attempting to evade rigorous scientific scrutiny in favor of a "fast-tracked" grant for a technically unstable architecture.
- D. Formal Challenge to Integrity:** The integrity of the public record has been formally challenged. Until the "Certificate of Integrity" and the "Fractured Record Notice" are resolved, any grant of this license would be a violation of the Administrative Procedure Act (APA). The Petitioner concurs that the Commission cannot move forward on a record that is mathematically and procedurally broken.
- E. "Prejudgment" of the Merits:** The Petitioner highlights a documented pattern of "Prejudgment" by the Commission's Chairman, Brendan Carr, regarding this specific proceeding. On February 4, 2026, the Chairman publicly endorsed the Applicant's "Kardashev II" narrative on social media before the close of the evidentiary record. Further, on March 11, 2026, the Chairman publicly scolded another Petitioner (Amazon) for filing an objection, instructing them to "focus on their own milestones" rather than the merits of the current application. These statements constitute a formal **prejudgment of the merits** and a departure from the role of an impartial adjudicator, creating a reversible error under the Administrative Procedure Act (APA).
- F. The "Platform Bias" Factor:** The Petitioner notes that the Chairman's prejudicial statements were issued exclusively on a private social media platform (X) owned and controlled by the principal (Elon Musk) of the Applicant (SpaceX). This creates an untenable "Platform Bias" where the venue for official regulatory commentary is owned by the party seeking the license. This environment facilitates an illegal ex parte atmosphere where the Applicant's narrative is prioritized over the verified technical data provided by independent Petitioners.

IX. Conclusion:

I must reiterate that under standard administrative procedure, the Applicant has effectively conceded all points in Section VI as uncontested facts. By failing to provide a single word of rebuttal to the fundamental thermodynamic, logistical, and biological hazards identified by the Petitioner, and corroborated by international experts, the Applicant has admitted to the technical insolvency of the

Gen3 architecture. SpaceX's silence on these thermal constraints is telling; they have proposed a system that requires "Kardashev II" energy output without even providing a "Type I" engineering solution for heat (the 2nd law of thermodynamics.) It does not address even the most basic laws of physics. SpaceX isn't just ignoring petitioners; they are trying to ignore the fundamental rules of the universe. To suggest that a '3-Kelvin refrigerator' exists in a vacuum is not an 'ambitious' engineering claim; it is a divorce from physical reality designed to protect a \$750B narrative, reflecting a level of hubris and technical oversight that borders on engineering negligence. Rendering a 'Public Interest' finding by this Commission would constitute a legal and physical impossibility.

These insurmountable physical constraints prove the Gen3 system is a thermodynamic impossibility. SpaceX's entire "Executive Summary" (Page 2 of their filing) is nothing more than a fairy tale; it is speculative to the point of being Science Fiction rather than validated scientific engineering fact (e.g., claiming 'abundant radiative cooling' without addressing the Stefan-Boltzmann constraints of the Radiator Paradox).

The Applicant demonstrates a systemic refusal to engage with the unique physical constraints of each orbital layer; SpaceX is trying to hide from the laws of physics. The Applicant's silence is not a defense; it is a confession of technical insolvency. SpaceX is "bankrupt" when it comes to the engineering required to make this work. Their entire document is a marketing brochure, not a technical opposition.

The FCC must recognize that this is not merely a technical disagreement, but a matter of regulatory integrity. Granting a "Novelty" waiver for a system that is physically incapable of operating, and which ignores established Prior Art, would effectively validate a **\$750B valuation "pump"** built on a foundation of engineering negligence. It is a request to approve a design that is **negligent, not ambitious**. The FCC cannot grant a waiver for a system that is physically incapable of operating without immediate catastrophic failure.

CERTIFICATE OF SERVICE

I hereby certify that on this 23rd day of March, 2026, a true and correct copy of the foregoing **Reply to Consolidated Opposition** was served via electronic mail upon the following counsel for Space Exploration Holdings, LLC:

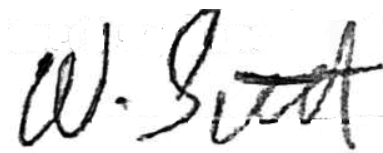
HWG LLP
William M. Wiltshire
Email: wwiltshire@hwglaw.com

Gibson, Dunn & Crutcher LLP
H. Mark Lyon
Email: mlyon@gibsondunn.com

Space Exploration Holdings, LLC
Space Exploration Technologies Corp.
1 Rocket Road
Hawthorne, CA 90250

Attn:
David Goldman, Vice President, Satellite Policy
Jameson Dempsey, Director, Satellite Policy
Cecilia Tenge-Rietberg, Senior Satellite Policy Manager

Emails: david.goldman@spacex.com
jameson.dempsey@spacex.com
cecilia.tenge-rietberg@spacex.com

A handwritten signature in black ink, appearing to read 'W. Stewart', written over a faint horizontal line.

/s/ William Stewart,
Petitioner